

Amy Anderson

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EDUCATION

Ph.D., Florida State University, Summer 2025

Dissertation: Applications of the Reduced Basis Method in Nuclear Physics.

Advisor: Dr. Jorge Piekarewicz

M.S., Florida State University, Fall 2021

M.S., Rice University, Fall 2016

B.A., Austin College, Spring 2015

PUBLICATIONS

“Universal reduced basis for the calibration of covariant energy density functionals.” *Physical Review C* 110.4 (2024): 044309.

“Applications of Reduced-Basis Methods to the Nuclear Single-Particle Spectrum.” *Physical Review C*, vol. 106, no. 3, 30 Sept. 2022, <https://doi.org/10.1103/PhysRevC.106.L031302>.

Space Dictionary for Kids: The Everything Guide for Kids Who Love Space. Prufrock Press Inc., 2016.

SKILLS

Programming: Python (primary), MATLAB, Mathematica, C++

Numerical Methods: Differential Equations solvers, reduced basis methods

Scientific Computing: Simulations, data analysis

RESEARCH EXPERIENCE

Analysis of Neutron Stars using Reduced Basis Methods (Fall 2022 – Present).

Collaborators: Dr. Pablo Guiliani, Dr. Alexandra Semposki, Dr. Sudhanva Lalit

Attempting to use a reduced basis and Galerkin projection to quickly and accurately analyze an equation of state for neutron stars.

Applications of Reduced Basis Methods to the Nuclear Single Particle Spectrum (Summer 2022 – Spring 2025).

Advisor: Dr. Jorge Piekarewicz,

Developed reduced basis method emulators to quickly and accurately calculate the energies and wavefunctions of all bound particle states in a spherical nucleus. Two papers have been published, concerning non-relativistic and relativistic calculations.

BAND Fellowship (Summer 2023). Competitive fellowship supporting nuclear physics research. Traveled to FRIB to collaborate on developing model order reduction software for compact astronomical objects, working primarily with Dr. Pablo Guiliani.

Simulating Particle Dynamics in Saturn’s F Ring (Summer 2014). NSF Research Experience for Undergraduates. Working with Dr. Lorin Matthews at Baylor University, used MATLAB to simulate the forces acting on charged dust grains in Saturn’s F ring.

Photometric Analysis of DQ Tau at Periastron (Summer 2016 – Fall 2016). Detailed observations of the binary T-tauri star DQ Tau used to examine the star's light curve and periastron brightening. This work was done under the supervision of Dr. Christopher Johns-Krull at Rice University.

Identifying a Natural Coordinate System for Investigating Gravitational Ripples in a Curved Spacetime (Fall 2014 – Spring 2015). Honors Thesis in Physics under Dr. Don Salisbury examining the curvature of spacetime in a region characterized by two interacting plane waves.

TEACHING EXPERIENCE

Florida State University TA: Four semesters of introductory physics, including both algebra and calculus-based. Experience both running a lab and facilitating recitation work

Varsity Tutors (Summer 2025 – Present) Online tutoring in math and physics at both the high school and college level

SAT/ACT Tutor (Summer 2018 – Summer 2019) Worked with students to prepare for all sections of the SAT and ACT standardized exams.

Physics Tutor (Spring 2017 – Summer 2019) Online tutoring helping students with homework and test preparation in physics and astronomy

PRESENTATIONS

Machine Learning Emulators for Nuclear Physics, Florida State University (Spring 2025)

Ghost in the Machine (Learning), Florida State University (Spring 2024)

Emulating the Single Particle Spectrum, Florida State University (Fall 2023)

Identifying a Natural Coordinate System for Investigating Gravitational Ripples in a Curved Spacetime, Austin College (Spring 2015)

Machine Learning and Me, Conference for the Advancement of Science Teaching (CAST) (Fall 2025).

Choose Your Own Adventure: Stellar Evolution, Conference for the Advancement of Science Teaching (CAST) (Fall 2025).